

Green Fuel Intelligence System (GFIS)

A Physics-Guided Hybrid AI Digital Twin for Methane Yield Prediction, Soft Sensing and Stability-Aware Simulation in Anaerobic Digestion

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Introduction & Problem Statement

Anaerobic digestion (AD) is one of the most intricate biological processes harnessed for renewable energy production. Microbial communities break down complex organic matter — livestock manure, food waste, agricultural residues — in the complete absence of oxygen to yield methane-rich biogas. The process unfolds across four biochemical stages: hydrolysis, acidogenesis, acetogenesis, and methanogenesis, each governed by distinct microbial consortia and sensitive to a cascade of interacting environmental variables.

Interacting Variables

Temperature, pH, organic loading rate (OLR), and volatile solids (VS) interact nonlinearly, making deterministic control extremely challenging without continuous monitoring.

Infrastructure Gap

Small and medium biogas plants frequently lack advanced SCADA infrastructure or complete laboratory instrumentation, creating significant blind spots in process visibility.

Stability Detection

Without real-time monitoring of chemical intermediates like VFAs and alkalinity, early detection of biological instability — a precursor to catastrophic reactor failure — is nearly impossible.

The consequences of undetected instability range from reduced biogas yields to complete methanogenic inhibition, representing significant economic losses and operational downtime for plant operators. GFIS was designed specifically to close this instrumentation and intelligence gap.

The Mechanistic vs. Black-Box Dilemma

The modeling landscape for anaerobic digestion has historically been divided between two fundamentally incompatible paradigms, each with critical limitations that prevent standalone deployment in industrial settings.

Traditional Mechanistic Models (ADM1)

The IWA Anaerobic Digestion Model No. 1 (ADM1) provides a scientifically rigorous, biochemically grounded representation of the digestion process. It encodes mass balances, enzymatic kinetics, and ionic equilibria with high fidelity. However, this expressiveness comes at a cost: ADM1 requires calibration of over 100 parameters, making real-time deployment computationally prohibitive and practically infeasible for field-scale plants without dedicated simulation infrastructure.

Standard Data-Driven Machine Learning

Purely black-box ML models — gradient boosters, deep neural networks — excel at learning complex nonlinear relationships from historical process data. They are fast to evaluate at inference time and require no explicit biochemical knowledge. The critical flaw: when extrapolating beyond training data distributions, they routinely generate thermodynamically impossible predictions such as methane yields exceeding the stoichiometric limit of the available substrate — a fatal flaw in safety-critical process environments.

- ❏ The industrial imperative demands a modeling framework that is simultaneously computationally practical for real-time deployment, statistically accurate under process variability, and biochemically reliable — never violating the physical laws governing microbial metabolism. This is the precise gap GFIS is engineered to fill.

Faculty Evaluation & The PIML Pivot

A pivotal evolution in the GFIS development cycle occurred following structured faculty evaluation. The review process identified a fundamental opportunity to elevate the framework beyond conventional black-box machine learning by introducing rigorous physical constraints directly into the learning objective.

Evaluator Recognition

The evaluating faculty recognized the latent value in exploiting the relationship between easily measurable online variables — temperature and pH — and difficult-to-measure chemical states such as VFA concentration and alkalinity. This insight catalyzed the pivot toward a physics-constrained modeling paradigm.

The PIML Transition

The project successfully transitioned from standard black-box ML to a Physics-Informed Machine Learning (PIML) approach. Rather than treating thermodynamic constraints as post-hoc guardrails, the PIML framework embeds them directly into the loss function, forcing the neural network to internalize physical feasibility as a core optimization objective during training.

Mathematical Enforcement

The loss function now actively penalizes predictions that violate biochemical possibility. Any estimated methane yield that exceeds the stoichiometrically permissible maximum — defined by the available volatile solids and ambient process conditions — incurs a severe quadratic penalty, ensuring the AI operates within the envelope of real-world chemistry at all times.

Current Scope and Objectives

The current GFIS iteration represents a fully functional, working research prototype deployed as a digital-twin-ready simulation dashboard. The system is architecturally prepared for real-time industrial deployment while currently being validated in a controlled research environment.



Feature Engineering Pipeline

A complete preprocessing pipeline handles missing values via forward-fill and median imputation, min-max normalization of sensor signals, construction of lag features capturing process memory, and rolling statistical aggregates reflecting biological inertia.



VFA/ALK Soft Sensing

Virtual soft sensing of the VFA/Alkalinity ratio replaces costly offline laboratory titration with near-real-time inference from online measurements, providing continuous visibility into process stability.



Methane Yield Prediction

The primary predictive objective — methane yield in Nm^3/day — is estimated through the physics-guided ensemble, with all predictions audited against thermodynamic feasibility bounds before being surfaced to operators.



Stability Classification

A three-state classification engine (Stable / Warning / Critical) synthesizes prediction outputs and soft-sensor signals into actionable plant-operator alerts, enabling proactive rather than reactive process management.

Validation is currently conducted on a carefully prepared synthetic and literature-inspired anaerobic digestion dataset, designed to replicate the statistical properties and process dynamics observed in real-world biogas plant operations while allowing controlled evaluation of edge-case scenarios.

Hybrid AI Architecture

GFIS is powered by a purpose-engineered hybrid AI architecture that strategically combines two complementary learning paradigms. Neither XGBoost nor LSTM alone would be sufficient for the demands of anaerobic digestion modeling — the architecture exploits the distinct strengths of each branch in a synergistic ensemble.

XGBoost Branch — Spatial Feature Intelligence

The XGBoost module ingests the current time-step feature vector and constructs an ensemble of gradient-boosted decision trees optimized for tabular nonlinear regression. It is particularly effective at capturing immediate kinetic interactions — such as the combined effect of a simultaneous pH drop and OLR surge — that manifest instantaneously in process measurements. XGBoost's inductive bias toward feature interaction modeling makes it the dominant branch for capturing sharp, nonlinear threshold effects in process chemistry.

- Gradient-boosted ensemble over tabular process features
- Captures immediate spatial feature interactions
- Robust to missing values and feature scaling
- Low inference latency — suitable for real-time evaluation

PyTorch LSTM Branch — Temporal Memory

The Long Short-Term Memory (LSTM) recurrent network processes a sliding window of historical process states, learning to recognize temporal patterns in reactor dynamics that unfold over hours to days. Biological processes like methanogenesis exhibit significant inertia — the effects of a feed overload may not manifest in gas production for 12 to 24 hours. The LSTM branch encodes this biological memory, capturing lag effects and trend dynamics that instantaneous snapshot models fundamentally cannot represent.

- Recurrent architecture retaining long-term sequence context
- Models biological delays and reactor inertia patterns
- Hidden state encodes accumulated process history
- Essential for early warning of slow-onset instabilities

The Ensemble Formulation

The two model branches are not evaluated independently — their outputs are combined through a learned, optimized ensemble formulation that produces a single, authoritative prediction at each time step. This weighted combination is the mathematical heart of the GFIS prediction engine.

Ensemble Prediction Equation

$$\hat{Y}_t = w_1 \cdot Y_{LSTM,t} + w_2 \cdot Y_{XGB,t} \quad \text{subject to} \quad w_1 + w_2 = 1.0$$

w_1 — LSTM Weight

The weight assigned to the LSTM temporal prediction. Optimized via cross-validation across multiple train/test splits to reflect the relative contribution of historical sequence context to overall predictive accuracy.

w_2 — XGBoost Weight

The weight assigned to the XGBoost spatial prediction. Its magnitude reflects the proportion of predictive variance explained by instantaneous feature interactions versus accumulated process history.

Optimization Constraint

The constraint $w_1 + w_2 = 1.0$ ensures the ensemble output remains in the same scale as the individual model predictions, preserving interpretability of the methane yield estimate in Nm^3/day .

Weights are determined through nested cross-validation on the training dataset, evaluating RMSE at each candidate weight combination over a fine grid. This ensures that the ensemble allocation reflects genuine predictive contribution from each branch rather than being arbitrarily prescribed. The resulting weighted average benefits from variance reduction through model diversity — the two branches make independent errors that partially cancel in the combined output.

The Need for Physics Constraints

A fundamental limitation of standard machine learning optimization must be confronted directly: conventional Mean Squared Error (MSE) minimization is entirely indifferent to physical feasibility. The optimizer only cares about minimizing the numerical discrepancy between predictions and training targets — it has no awareness of thermodynamics, stoichiometry, or biochemical possibility.

1

Standard MSE Objective

Minimizes $\sum(\hat{y} - y)^2$ purely on statistical grounds. A prediction of 200 Nm³/day from a reactor with VS capable of yielding only 80 Nm³/day is numerically penalized only by its distance from the training label — the thermodynamic impossibility is invisible to the loss function.

2

The Fatal Extrapolation Flaw

When the model encounters an operating condition outside its training distribution — a novel feed composition or an extreme temperature excursion — it may generate methane yield predictions that exceed the stoichiometric limit of the substrate. In plant operations, acting on such predictions leads to dangerous misallocations of resources and missed alarm conditions.

3

Physics-Constrained Enforcement

GFIS departs entirely from pure MSE optimization. The custom physics-guided loss function introduces a hard biochemical ceiling derived from simplified ADM1 stoichiometry, actively penalizing the network whenever it attempts to generate a prediction that the substrate chemistry cannot support.

Mathematical Foundation of PIML

The Physics-Informed Machine Learning core of GFIS is implemented through a custom loss function that augments standard data-fidelity error with a thermodynamic feasibility penalty term. This formulation forces the neural network to simultaneously minimize prediction error and respect substrate-derived yield limits — the two objectives are jointly optimized during every gradient update.

Physics-Guided Loss Function

$$L_{PG} = \underbrace{MSE(Data)}_{\text{Data Fidelity}} + \lambda \cdot \underbrace{\max(0, \hat{y}_t - F_{max}(VS_t))^2}_{\text{Thermodynamic Penalty}}$$

Term Definitions

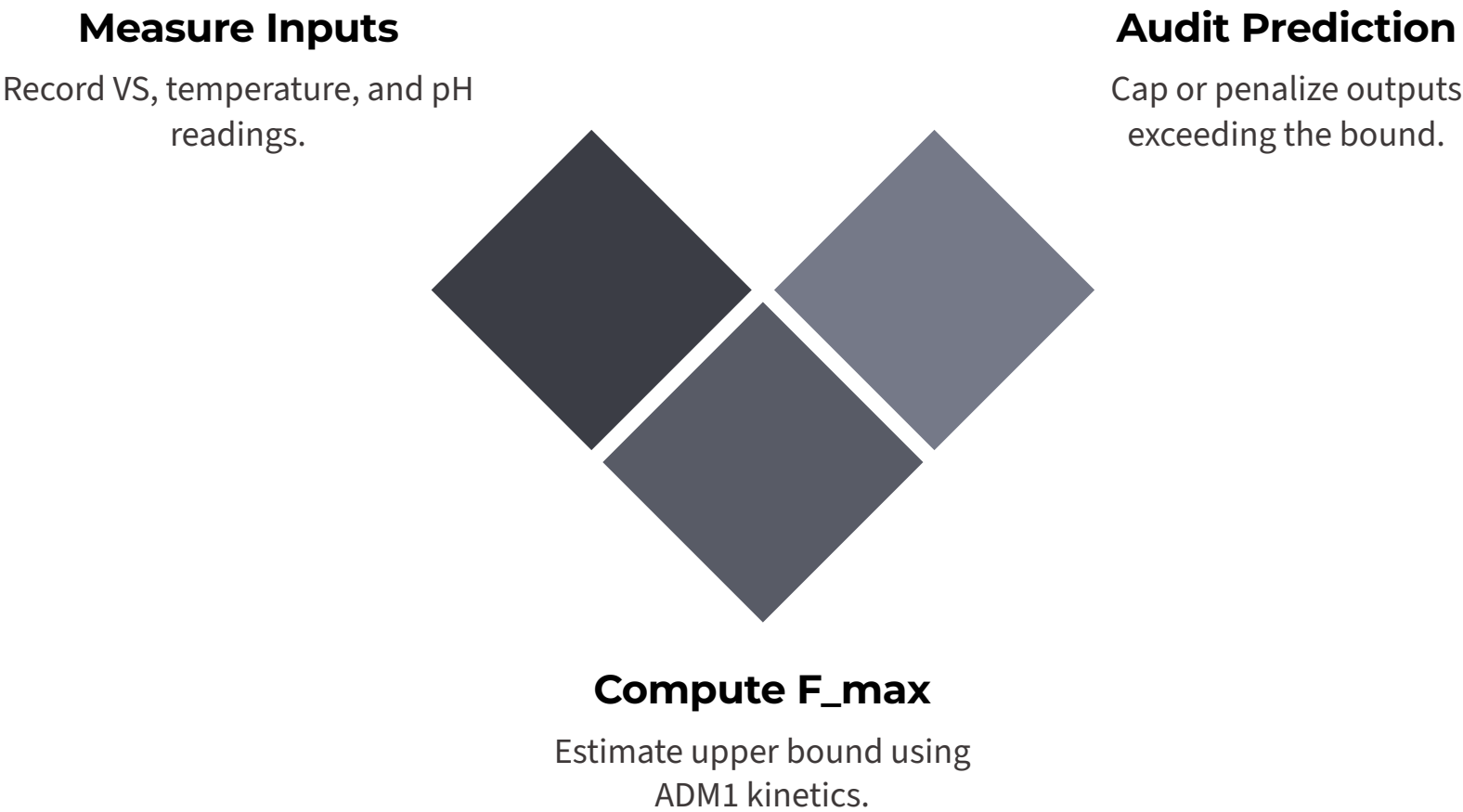
- **MSE(Data):** Standard mean squared error between predicted methane yield $\hat{y}_t$ and observed target y_t , ensuring statistical accuracy on the training data.
- $F_{max}(VS_t)$: The maximum thermodynamically feasible methane yield at time t , computed dynamically from the available volatile solids concentration and ambient pH and temperature conditions via simplified ADM1 kinetic relations.
- λ : A regularization hyperparameter controlling the severity of the physics penalty. Tuned via validation to enforce feasibility without suppressing legitimate high-yield predictions.
- $\max(0, \cdot)^2$: The hinge-squared formulation ensures the penalty activates only when the prediction exceeds the physical bound — predictions below the ceiling are not penalized, preserving statistical accuracy for feasible outputs.

Gradient Behavior

The hinge-squared penalty produces a smooth, differentiable gradient when active, making it fully compatible with PyTorch's automatic differentiation engine. When $\hat{y}_t \leq F_{max}$, the penalty gradient is zero and the network optimizes purely for data fidelity. When $\hat{y}_t > F_{max}$, the gradient magnitude grows proportionally to the degree of violation, creating a strong restoring force that pushes the network weights away from the infeasible regime. This asymmetric loss landscape is precisely what enforces the biochemical ceiling without constraining the model in the feasible region.

Implementing the Feasibility Bound

The physics layer of GFIS functions as a dynamic biochemical guardrail — an auditing mechanism that evaluates every prediction against a substrate-derived upper limit computed in real time from available process measurements. This bound is not static; it adapts continuously as feed composition, temperature, and pH evolve during plant operation.



The maximum physically permissible methane yield $F_{max}(VS_t)$ is computed using a simplified Buswell-Boyle stoichiometric relation scaled by Monod-type temperature and pH activity functions. These activity multipliers reduce the effective ceiling when conditions are suboptimal — for instance, a low pH environment suppresses methanogenic activity, lowering the realistically achievable yield even when volatile solids are abundant. This means the feasibility bound captures not just substrate availability but also the microbial capacity to utilize that substrate under prevailing conditions.

Volatile Solids Basis

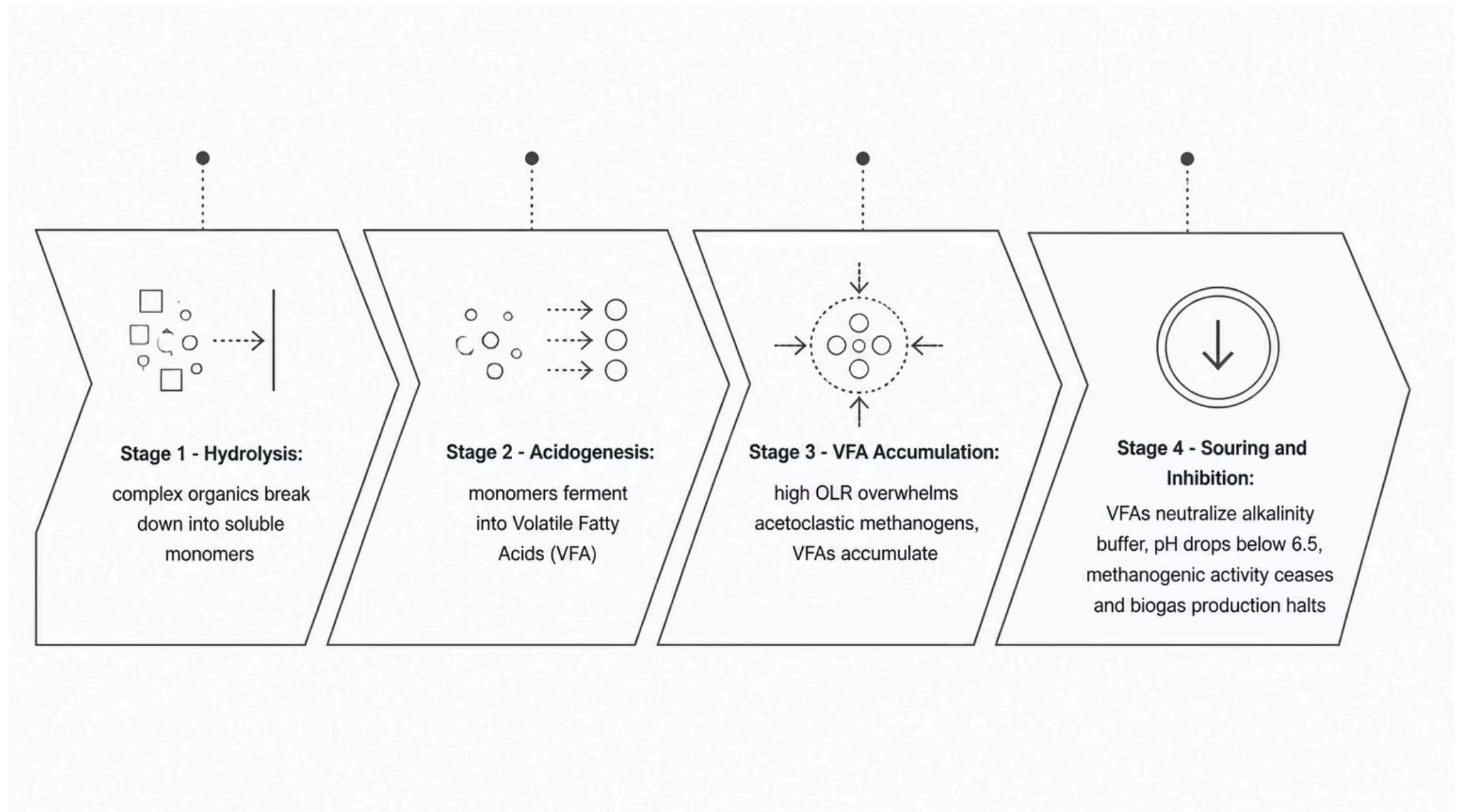
The substrate ceiling scales linearly with VS concentration. Higher organic loading provides more degradable carbon, increasing the theoretical methane potential — but this ceiling is always finite and measurable, providing a hard upper limit that the model cannot exceed.

Temperature & pH Activity Scaling

Methanogenic microorganisms have a narrow optimal operating window (35–38°C for mesophilic, pH 6.8–7.4). The activity multipliers programmatically reduce the effective yield ceiling when conditions deviate from this optimal window, tightening the constraint in hostile process environments.

Biological Reality: The Souring Pathway

One of the most significant failure modes in anaerobic digestion — and a central concern that GFIS was specifically designed to detect and prevent — is **Reactor Souring**. This cascade failure begins silently, often appearing benign in gas production metrics until irreversible damage to the methanogenic community has occurred.



The Trigger Mechanism

Souring is initiated when the organic loading rate exceeds the metabolic capacity of the syntrophic acetogen and methanogen populations. Hydrolysis and acidogenesis proceed at their natural rates, but acetoclastic methanogens — the microbes responsible for converting acetate into methane — become the rate-limiting bottleneck. VFAs therefore accumulate in the liquid phase faster than they are consumed.

The Irreversibility Threshold

When accumulated VFAs overwhelm the bicarbonate alkalinity buffer and pH drops below 6.5, methanogenic microorganisms — which are highly pH-sensitive and slow-growing — begin to lyse. Recovery from advanced souring requires weeks of remediation including alkalinity supplementation, feed cessation, and controlled inoculum reintroduction. Early detection via the VFA/ALK ratio, as implemented in GFIS, is the only practical prevention strategy.

The Virtual Soft Sensor

At the core of the GFIS stability intelligence layer lies the Virtual Soft Sensor — an XGBoost-based inference engine that eliminates the most critical instrumentation bottleneck in biogas plant operations: the latent, costly measurement of Volatile Fatty Acids and Alkalinity.

The Measurement Problem

Direct physicochemical measurement of VFA concentration and alkalinity requires off-line laboratory titration on extracted liquid samples — a process that introduces delays of 4 to 24 hours between sampling and result availability. During this latency window, a rapidly souring reactor can cross the irreversibility threshold. Furthermore, the analytical equipment, reagents, and skilled laboratory personnel required represent significant ongoing operational costs that small-to-medium biogas plants cannot sustain.

The Soft Sensor Solution

The VFA/ALK Virtual Soft Sensor infers this difficult-to-measure stability ratio from variables that are continuously and inexpensively available online: reactor temperature, liquid pH, feed organic loading rate, and biogas flow rate. XGBoost is the optimal model architecture for this task — its gradient-boosted tree ensemble excels at learning the complex, threshold-dominated relationships between these proxy measurements and the underlying chemical state of the reactor liquor.

4–24h

Lab Titration Delay

Typical latency between liquid sampling and VFA/ALK result availability in conventional plant operations.

~0s

Soft Sensor Latency

Near-instantaneous inference from continuously streaming online sensor measurements — no laboratory involvement required.

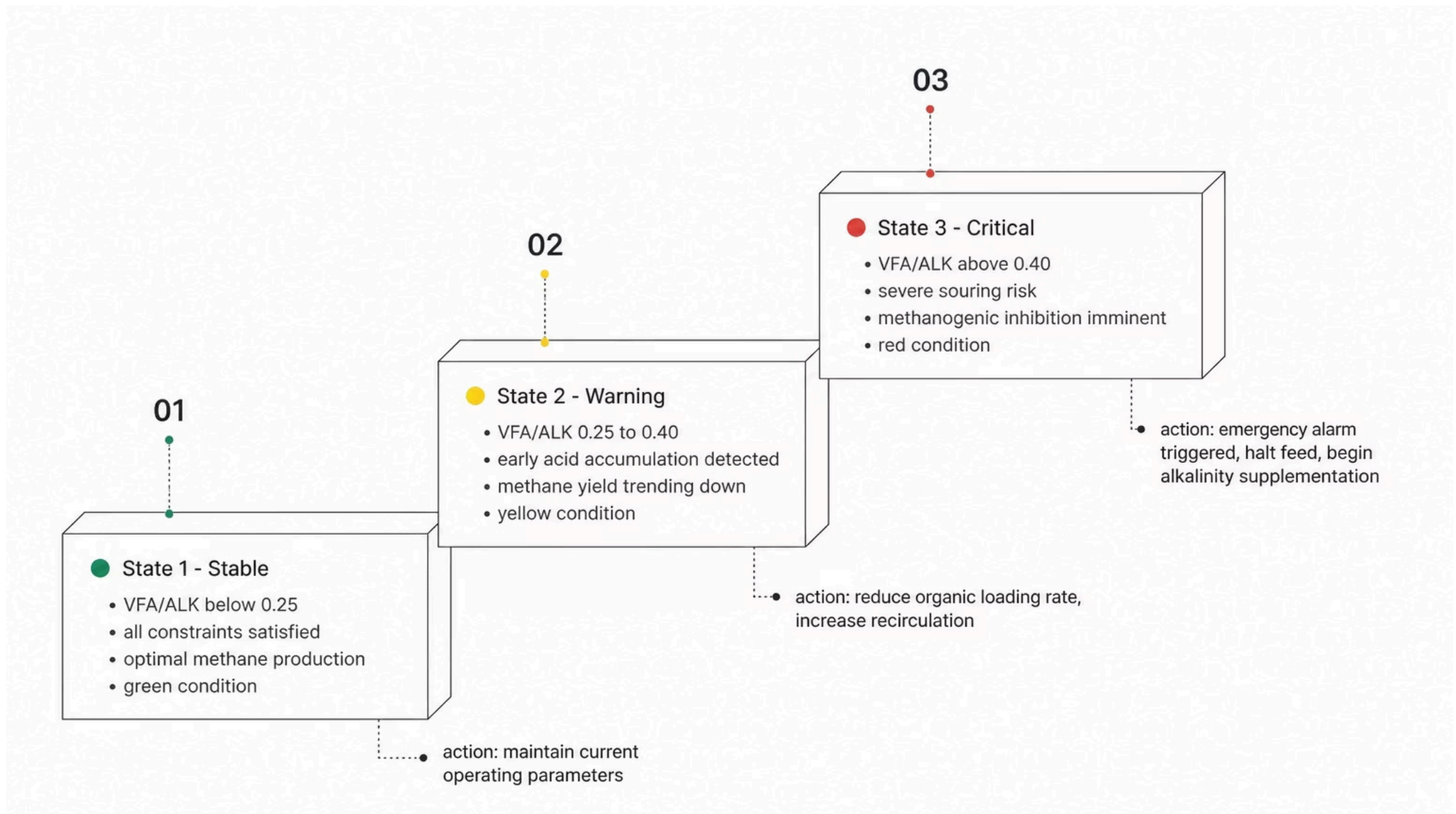
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Critical VFA/ALK Threshold

The ratio value above which GFIS triggers an emergency alarm indicating severe souring risk and potential reactor failure.

Stability Classification

The GFIS stability classification engine synthesizes outputs from all predictive and sensing modules — the physics-guided methane yield prediction, the thermodynamic feasibility audit, and the VFA/ALK soft sensor — into a single, actionable operational state that plant operators can act on immediately, without requiring deep process chemistry expertise.



The three-state classification framework was designed in direct consultation with process engineering literature on anaerobic digestion operational thresholds. The VFA/ALK ratio boundaries (0.25 and 0.40) are drawn from established consensus in biogas engineering practice: ratios below 0.25 indicate a healthy, well-buffered process; ratios between 0.25 and 0.40 signal developing imbalance that is still recoverable through operational adjustment; ratios exceeding 0.40 represent an acute crisis requiring immediate intervention. By mapping soft-sensor outputs to these operationally validated thresholds, GFIS provides recommendations that are directly actionable by plant technicians without requiring real-time laboratory confirmation.

Software Guardrails & Code Logic

The GFIS software architecture implements the biochemical constraints and stability logic through a structured Python module hierarchy. Each class encapsulates a distinct layer of process physics, ensuring that mathematical guardrails are systematically enforced throughout the prediction pipeline rather than being applied as ad-hoc post-processing corrections.

1

PhysicalConstraints Class

Executes the core biochemical mathematics. Computes the volatile-solids-based methane ceiling $F_{max}(VS_t)$ using temperature- and pH-adjusted activity functions. Programmatically models metabolic activity drop: if reactor pH falls outside the range 6.5–8.0, microbial activity is reduced by 80% in the constraint calculation, reflecting the physiological intolerance of methanogens to acidic or highly alkaline conditions.

2

PhysicsGuidedLoss Module (PyTorch)

A custom `nn.Module` implementing the PIML loss function using tensor operations. The hinge-squared penalty is computed via `torch.clamp(prediction - F_max, min=0) ** 2`, which is fully differentiable and integrates seamlessly with PyTorch's autograd engine for end-to-end backpropagation. The penalty coefficient λ is registered as a hyperparameter tunable via the training configuration.

3

SoftSensorMonitor Class

Processes biogas flow deviation signals, liquid pH readings, and OLR measurements to generate a continuous VFA/ALK estimate via the trained XGBoost soft sensor. Maps the estimated ratio against the three operational thresholds to output the final stability state (Stable / Warning / Critical) along with a structured alarm payload suitable for dashboard display and IoT event propagation.

The modular architecture ensures that each guardrail layer can be independently updated, recalibrated against new process data, or extended with additional biochemical constraints without requiring systemic refactoring — a critical design property for a system intended to evolve alongside the datasets and operating conditions it encounters in industrial deployment.

eXplainable AI (XAI) & SHAP Integration

The sophistication of a hybrid PIML architecture is valuable only if plant operators can trust and interpret its outputs. A prediction system that functions as an opaque black box — outputting a methane yield number or a "Warning" alarm with no attribution — will be rejected or overridden by experienced plant technicians who rely on their domain intuition. GFIS addresses this through full integration of SHAP (SHapley Additive exPlanations), transforming every model output into a traceable, parameter-attributed explanation.



SHAP Mathematical Basis

SHAP values are grounded in cooperative game theory. Each feature's contribution to a specific prediction is calculated as the weighted average of its marginal contributions across all possible feature coalitions. This produces a consistent, locally accurate decomposition of any model output into additive feature attributions that sum exactly to the prediction minus the baseline.



Actionable Industrial Insights

For every VFA/ALK warning event, GFIS generates a SHAP attribution report that quantifies the exact marginal contribution of each process variable — for example, identifying that a 3°C temperature drop contributed 0.08 VFA/ALK units to the warning trigger, while an OLR spike contributed 0.12 units. This transforms the alarm from a binary signal into an operational diagnosis.



Operator Trust & Adoption

By surfacing which parameters drove each prediction and alarm, SHAP integration enables plant operators to validate model reasoning against their domain knowledge, building trust through transparency. Operators can confirm that the model is reacting to genuinely alarming process conditions rather than spurious sensor noise, significantly increasing the likelihood of appropriate corrective action.

Preliminary Model Evaluation

Current performance metrics were derived from rigorous testing on the synthetic and literature-inspired anaerobic digestion dataset, prepared to replicate the statistical characteristics and process dynamics of real-world biogas plant operations. While these benchmarks will be updated upon ingestion of continuous industrial time-series data, they provide a meaningful baseline assessment of the system's predictive and constraint-enforcement capabilities.

XGBoost Tabular Branch

The XGBoost branch demonstrated strong performance on the spatial feature prediction task, achieving metrics that validate its ability to capture nonlinear kinetic interactions from instantaneous process snapshots.

- **R²:** 0.7556 — explaining 75.6% of methane yield variance from contemporaneous features
- **RMSE:** 6.7933 Nm³/day — root mean squared prediction error
- **MAE:** 5.5517 Nm³/day — mean absolute prediction error, indicating typical operational accuracy

LSTM Temporal Branch

The LSTM branch yields an initial R² of 0.2942 on the synthetic dataset. This lower metric is expected and explained: LSTM sequence models depend critically on long, continuous, high-frequency time-series data to learn meaningful temporal patterns. Synthetic datasets, by their construction, lack the authentic autocorrelation structures and process inertia signatures present in real reactor data streams. Performance is projected to improve substantially upon integration of continuous industrial sensor data.



Zero Physics Violations

The most critically important evaluation result: the physics-guided ensemble recorded **strictly zero methane-bound violations** during the entire evaluation dataset. No prediction exceeded the substrate-defined thermodynamic ceiling — confirming that the PIML loss enforcement operates correctly under the full range of test conditions.

Ensemble Calibration Path

The ensemble weights w_1 and w_2 will be re-optimized following industrial dataset integration, as the relative contribution of temporal versus spatial features may differ substantially between synthetic and real-world operating conditions.

Web-Based Simulation Dashboard

The GFIS mathematical and AI backend is surfaced to engineers and operators through a highly interactive web-based simulation dashboard, served via a FastAPI application backend with a purpose-designed HTML/CSS frontend. The interface was developed with an explicit design philosophy: it must feel like a cyber-physical industrial control interface, not a generic data analytics portal.

Interface Design Philosophy

The frontend intentionally avoids decorative graphics and animated distractions. It employs a high-contrast monochrome and orange aesthetic to render SCADA-style visualizers entirely in CSS, without external charting libraries — creating the visual grammar of industrial process control systems that engineers instinctively recognize and trust. Every panel maps to a specific predictive or monitoring function: methane yield forecast, VFA/ALK trend, stability state indicator, and SHAP attribution overlay.

What-If Simulation Engine

A central capability of the dashboard is its function as an accelerated scenario simulation engine. By decoupling computational time from physical time, operators can test hypothetical process perturbations — such as a 20% increase in organic loading rate, a feed temperature drop of 5°C, or a switch to a high-lipid feedstock — and observe the predicted system response instantaneously. This capability transforms the dashboard from a passive monitoring tool into an active process optimization instrument, enabling informed operational decisions before perturbations are physically introduced to the reactor.

The FastAPI backend exposes RESTful prediction endpoints, enabling straightforward integration with upstream IoT data pipelines and downstream alarm management systems. The API architecture is already designed for the sensor data ingestion patterns required for full digital twin operation, making the transition from simulation dashboard to live monitoring system an incremental engineering task rather than an architectural redesign.

Stress-Testing: Controlled Scenarios

The GFIS dashboard includes a purpose-built 48-hour controlled scenario simulator that subjects the physics-guided ensemble and soft sensor to adversarial operating conditions, evaluating system response beyond the nominal operating envelope. These stress tests are designed to validate that the model interprets the holistic chemical stability of the reactor — not merely pattern-matching surface-level gas production metrics.

OLR Overload Scenario

The organic loading rate is stepped to 150% of the nominal operating set point for a sustained 12-hour window, simulating a feed pump malfunction or intentional overloading. **Observation:** The predicted mean methane yield remained relatively stable during the initial overload phase — a superficially reassuring signal that a naïve monitoring system would report as normal operation. However, the VFA/ALK soft sensor detected the hidden acid accumulation signature building in the process chemistry, correctly classifying the reactor into the **"Warning"** state approximately 8 hours before the gas production decline became apparent in the yield metric. This demonstrates the soft sensor's critical early-warning value beyond what yield monitoring alone can provide.

Corrective Recovery Scenario

Following the simulated OLR overload, corrective actions are modeled: organic loading is reduced to 60% of nominal, alkalinity supplementation is introduced, and the system is held at these adjusted conditions. **Observation:** The stability classifier progressively returns from "Warning" to "Stable" classification over a 14-hour recovery window, with the VFA/ALK ratio declining below the 0.25 threshold as the simulated alkalinity buffer is restored. This recovery trajectory closely mirrors published experimental observations of AD process recovery in the literature, providing indirect validation of the model's biological realism.

- ❏ These controlled scenarios provide qualitative validation evidence that GFIS captures genuine biochemical process dynamics rather than producing statistically fitted but physically meaningless predictions. The system's ability to issue early warnings during apparently stable gas production phases is its most operationally significant capability.

Industry Support & Project Milestones

The GFIS framework has achieved meaningful external validation beyond the academic research environment, demonstrating its viability as a production-grade industrial tool through competitive exhibition recognition and formal industry sponsorship.

Research Inception

GFIS concept formulated at BITS Pilani under M.Tech AI/ML program. Initial scope defined as physics-informed digital twin for anaerobic digestion process optimization.

DIPEX 2026 — First Prize

GFIS was presented at the **DIPEX 2026** technology exhibition in Aurangabad (Sambhaji Nagar), competing against projects from institutions and industry participants across the region. The framework was awarded **First Prize**, providing independent recognition of its technical novelty and practical engineering merit by evaluators from the academic and industrial communities.

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PIML Architecture Development

Following faculty evaluation, system transitioned from black-box ML to Physics-Informed ML paradigm. Physics-Guided Neural Network and VFA/ALK soft sensor implemented and validated on synthetic dataset.

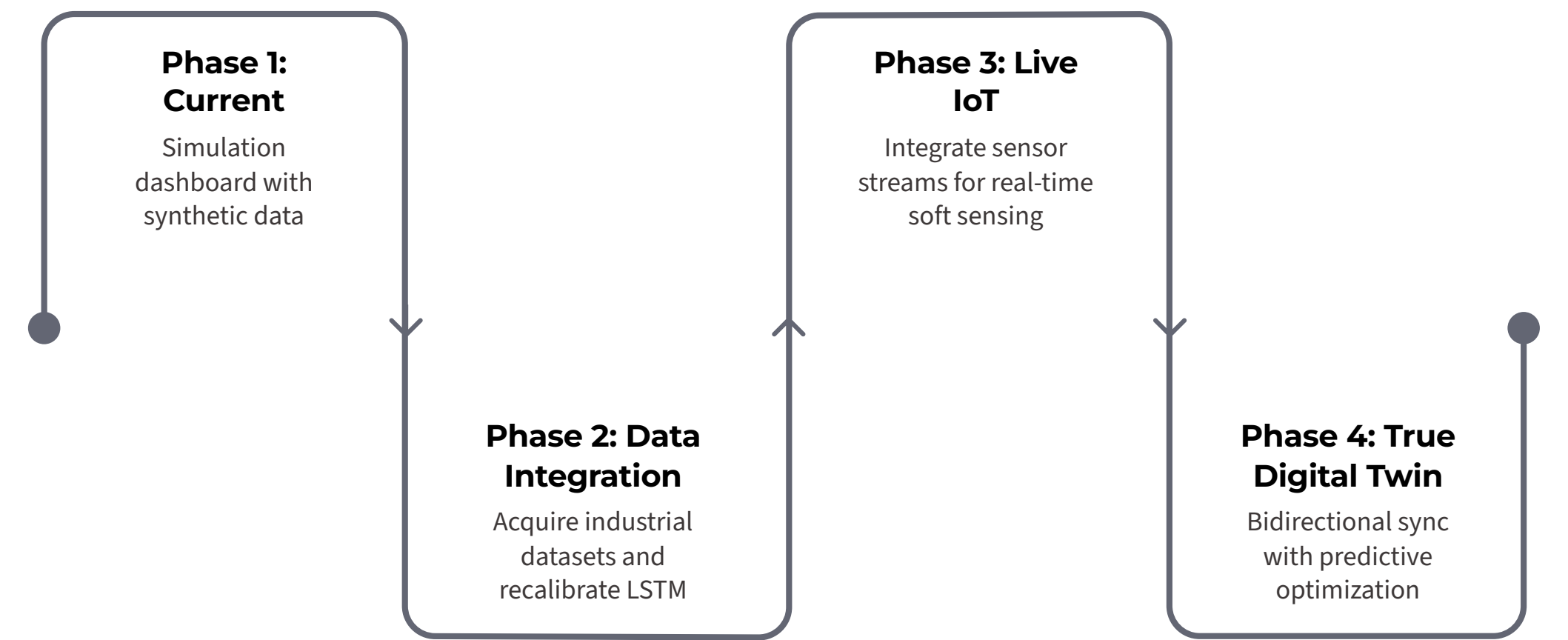
Chatake Innworks Sponsorship

Research, development, and testing activities are formally supported by **Chatake Innworks Pvt. Ltd.** as an independent industry sponsor. This sponsorship represents a commercial-sector assessment that GFIS possesses realistic production-deployment potential, distinguishing it from purely theoretical academic contributions.

The combination of competitive first-prize recognition and formal industry backing positions GFIS as a research output with a credible pathway from laboratory prototype to field-deployed industrial system — a transition that the framework's IoT-ready architecture and FastAPI backend are explicitly designed to support.

Future Roadmap to a True Digital Twin

As correctly identified during faculty evaluation, the current GFIS iteration is most precisely characterized as a **Physics-Guided Simulation Dashboard**. The transition to a true Digital Twin — an ISO 23247-compliant, bidirectionally synchronized virtual replica of a physical asset — requires the addition of continuous, real-time data ingestion from physical sensors operating on live industrial reactors. This transition is the primary objective of the next project phase.



Industrial Dataset Acquisition

The immediate priority is acquiring a high-quality, continuous industrial time-series dataset from an operating biogas plant — ideally at hourly resolution across Temperature, pH, OLR, VS, gas flow, and VFA/ALK measurements spanning at minimum 12 months of operation. This will enable LSTM recalibration, ensemble re-optimization, and statistically rigorous real-world validation.

IoT Sensor Integration

The GFIS IoT architecture is already designed for full-scale sensor integration. MQTT or OPC-UA protocol connectors will pipe live Temperature, pH, OLR, and VS measurements directly from field instruments into the prediction pipeline, replacing synthetic data ingestion with real-time physical sensor streams and completing the digital twin feedback loop.

Autonomous Control Advisory

The long-term research vision extends GFIS from passive monitoring and prediction to an active process optimization agent — generating real-time operating set-point recommendations (optimal OLR, pH correction dosing, temperature set-point) based on multi-step-ahead yield forecasting and stability risk quantification, bringing the system into the domain of model predictive control.

The GFIS framework represents a foundational contribution to the application of Physics-Informed Machine Learning in industrial bioprocess engineering — demonstrating that AI systems deployed in safety-critical biological environments need not choose between predictive power and physical reliability. The path to full digital twin deployment is clearly defined, architecturally prepared, and industry-supported.